Watermarks

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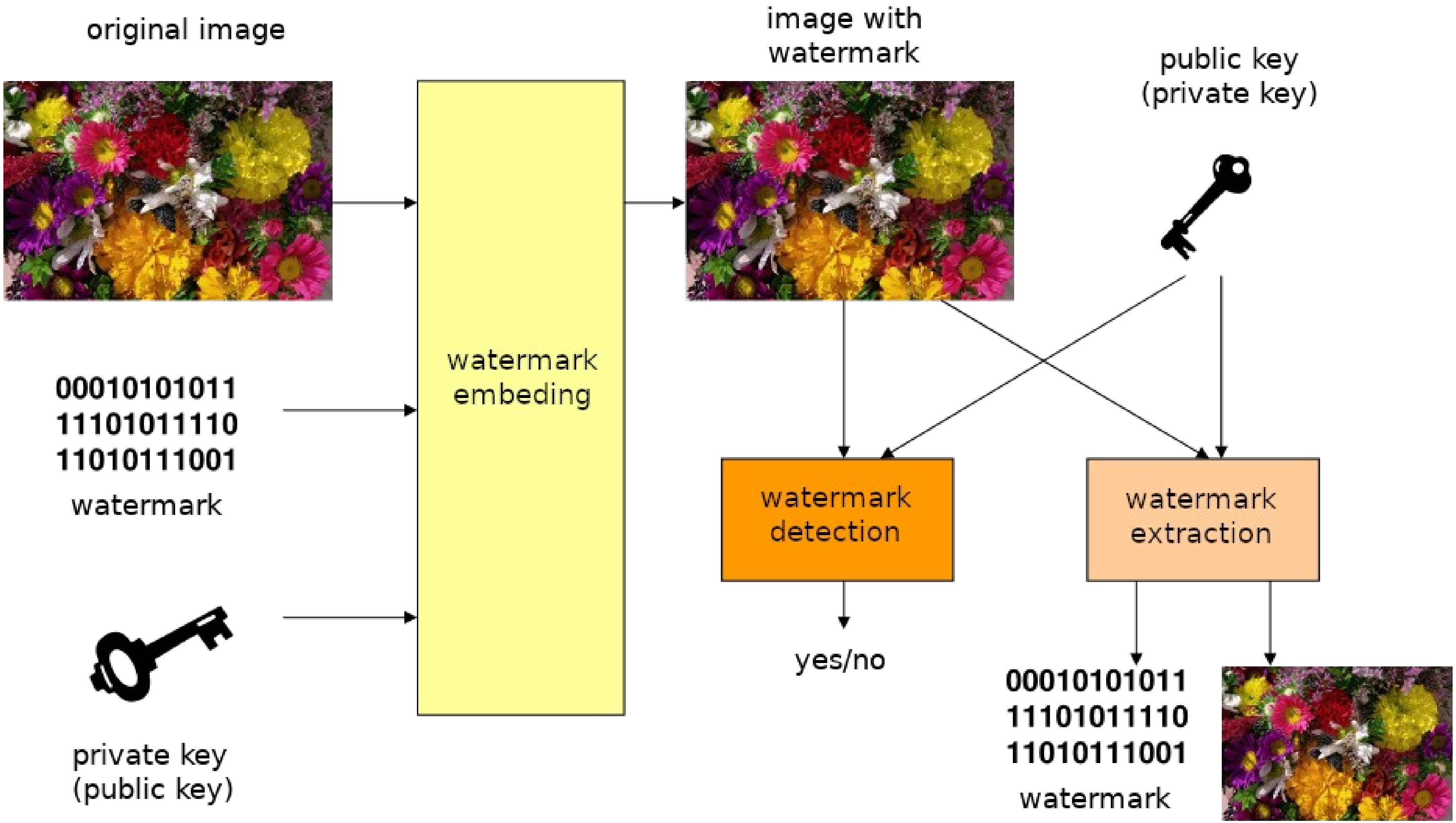
Outline

- Introduction
- Use cases / features of watermarks
- Methods
  - LSB Modulation
  - DCT coeficients
  - Disjoint sets
  - MD5
  - Visual Cryptography
  - Watermarks in text
- Attacks
  - Forensic analysis
- Summary

digital watermark

noun

a piece of code embedded in a digital image, video, or audio file in order to provide copyright information, typically being undetectable during normal use of the file.
"you might also consider adding a digital watermark to your photos, which renders them unusable for commercial purposes"


original image
image with
watermark
public key
(private key)
watermark
embeding
00010101011
11101011110
11010111001
watermark
watermark
detection
watermark
extraction
yes/no
private key
(public key)
00010101011
11101011110
11010111001
watermark

Classification

- Visible/invisible
- Method of insertion
  - Time domain – direct change of image
  - Frequency domain – e.g. changes of DCT coefficients during compression, change of spectrum, etc.
  - …
- Type of data
  - Binary – pseudo-random sequences, additional data
  - Image (logo) - from the extracted watermark can be visually derived to operations that were performed above the source image.
  - ...

Features

- Capacity
- Computational Complexity
- Granularity
- Invisibility
- Fragile/Robust
- Fault tolerance
- Security

Features

- Capacity (data payload) – how much information can watermark transfer (number of bits)
- Computational Complexity – Time/Space/Other, Real-time processing needs
- Granularity - minimal spatio-temporal interval for reliable insertion and/or detection
- Invisibility (fidelity) - Perceptibility
- Fragile/Robust
- Fault tolerance (false positive rate, MSE)
- Security
  - Unauthorized adding/removing/decoding/modification of watermark shouldn’t be possible
  - Kerckhoffs's principle:

    A cryptosystem should be secure even if everything about the system, except the key, is public knowledge.

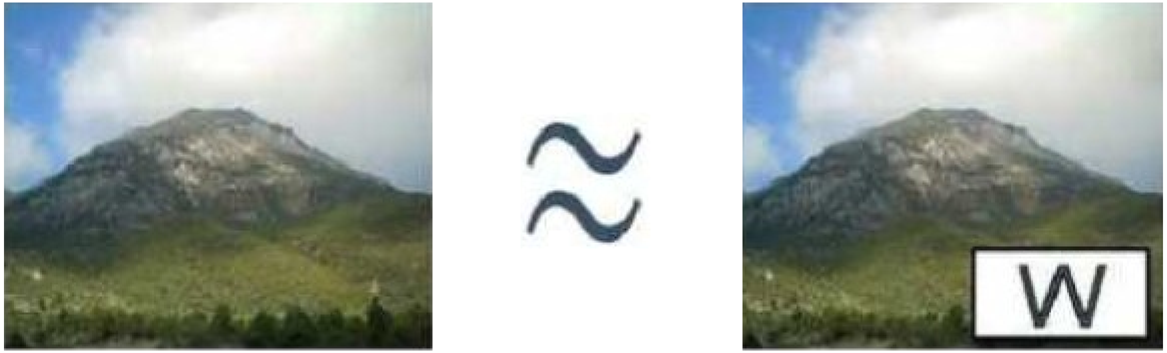
Visible watermarks

- Intentional degration of image quality by watermark
- Used in freely distributed frames, motion pictures, etc.
- Inserted information about author (e.g. his logo).
- Can cause a decrease in the quality of the image (for example: sub-sampling of the image).
- It is difficult to remove
  - De-Mark GAN [wu2018].
- It is possible to compromise the author by subverting a similar watermark on a false work.
- Suitable combination with invisible watermark.

Invisible watermarks

- The user will not recognize their presence in the tracked image during normal use.
- Common types
  - Fragile – to protect the integrity of the source data, the watermark is degraded during data transformation.
  - Robust - for copyright protection. It must be preserved after transformation operations (scaling, rotation, cropping, formatting, compression, printing / scanning, noise, ...).
- Robust watermarks
  - Private – for detection of watermark is necessary access to original data. This could cause problem when working with large data set.
  - Public – access to original data is not necessary. Proof of credibility usually requires detection, no extraction is necessary.

correspondence with original image

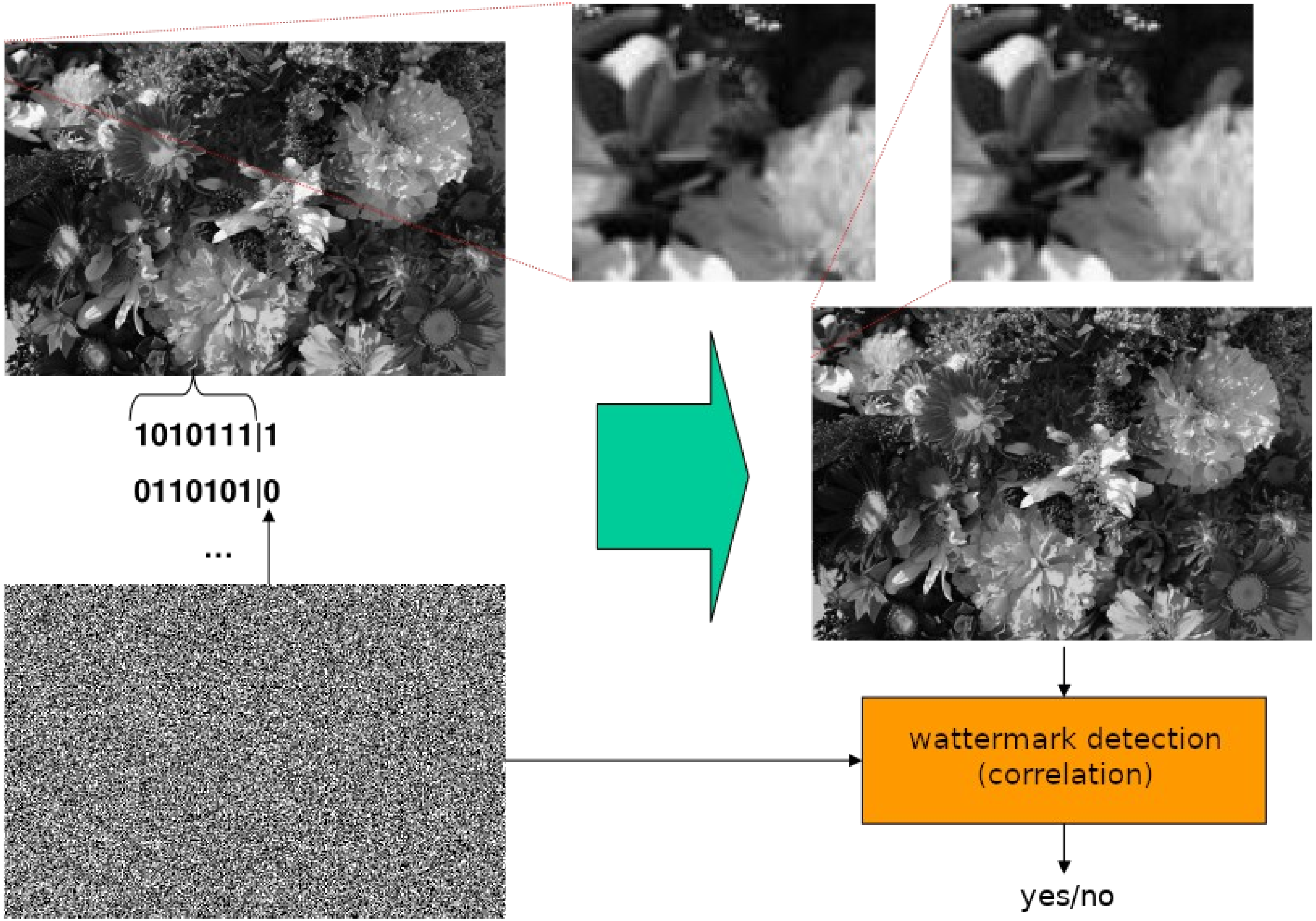


tolerance or sensitivity against common changes

LSB modulation

- Watermark is encoded to **least significant bits** of each pixel of image
- This kind of change is visually almost unrecognizable.
- Method **is not robust**. Watermark could be easily removed (e.g. by adding noise to least significant bit)
- The watermark could consist of pseudo-random sequence.
- Private key defines initial value of generator.
- Watermark could be detected using signal corelation.

LSB modulation



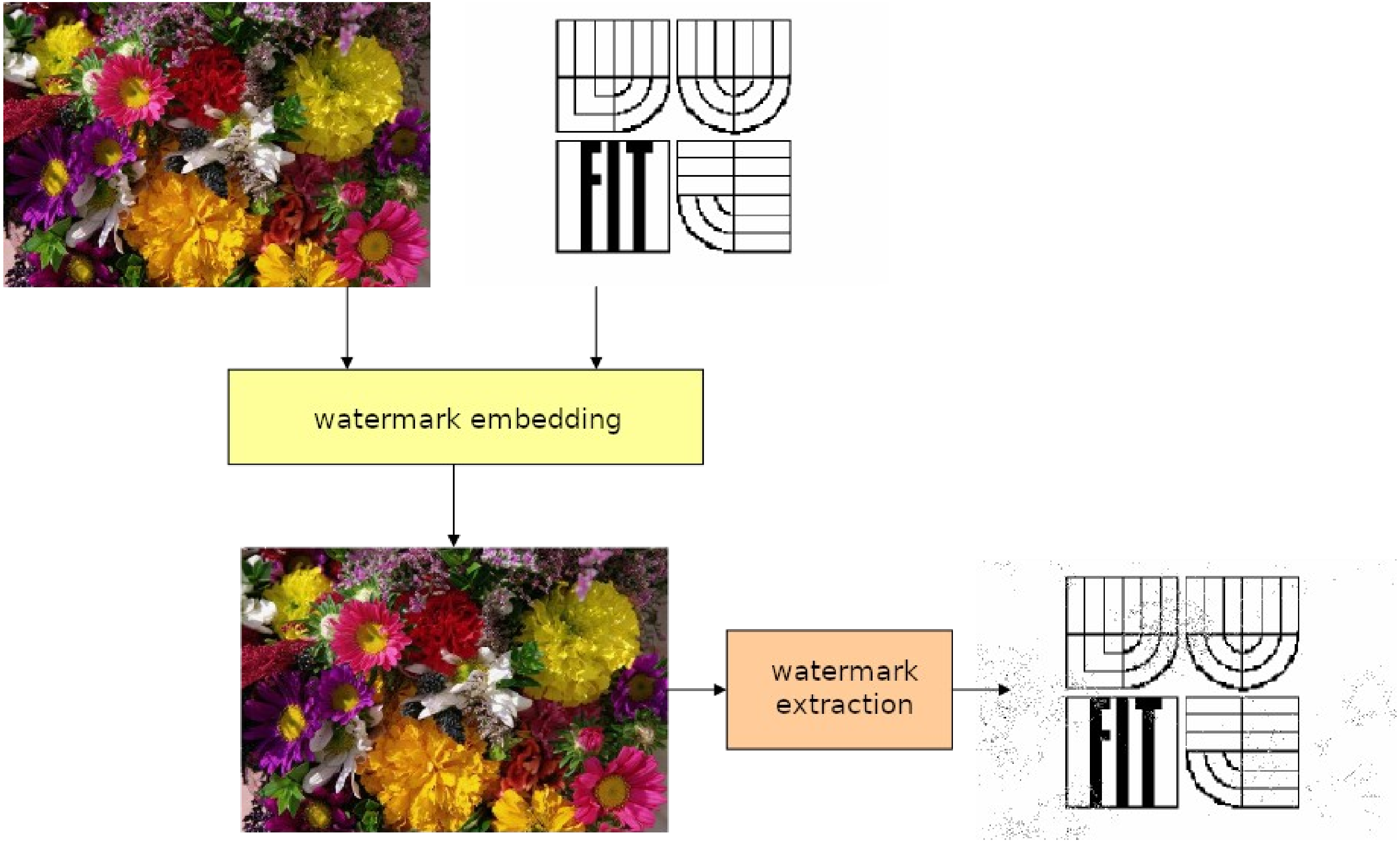
LSB modulation for color image

- Conversion into YUV color space (from RGB)
- Encode watermark into **Y channel**
- More robust to color changes
- More robust to compression (JPEG stores U and V channels sub sampled

$$\begin{bmatrix} Y' \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.14713 & -0.28886 & 0.436 \\ 0.615 & -0.51499 & -0.10001 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

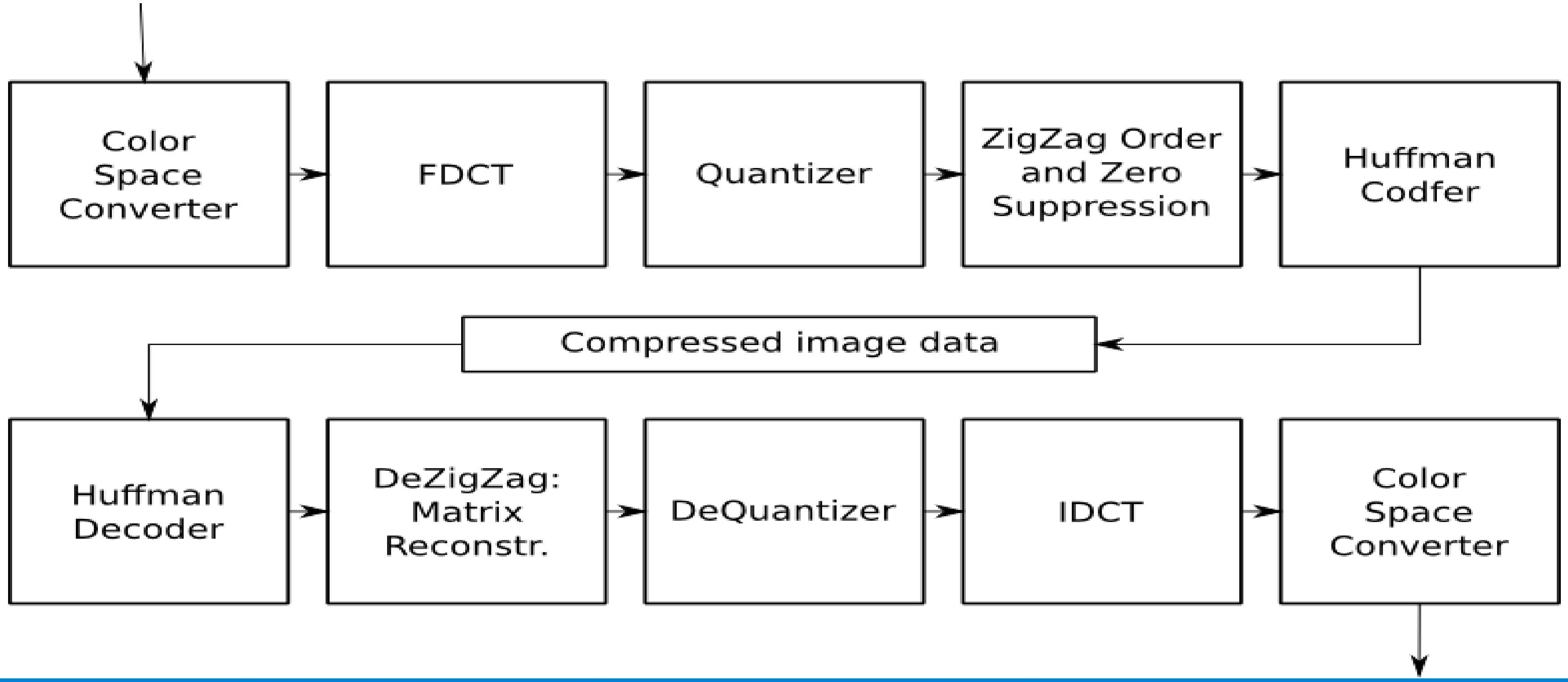
$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.13983 \\ 1 & -0.39465 & -0.58060 \\ 1 & 2.03211 & 0 \end{bmatrix} \begin{bmatrix} Y' \\ U \\ V \end{bmatrix}$$

LSB modulation for color image



Embedding of watermark using DCT coefficients

- LSB modulation in source image is not very robust against lossy compression (e.g. JPEG, MPEG)
- Wattermark disapears during one phase of compression algorithm



Embedding of watermark using DCT coefficients

- It is more appropriate to modify the coefficients obtained with DCT during compression.
- It is much more robust and less visible in output picture.
- It is necessary to tamper compression algorithm.

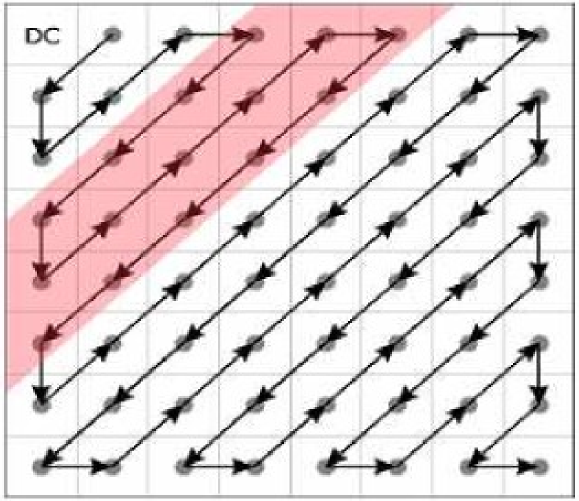
$$F(m,n) = \frac{2}{\sqrt{MN}} C(m) C(n) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cos \frac{(2x+1)m\pi}{2M} \cos \frac{(2y+1)n\pi}{2N}$$

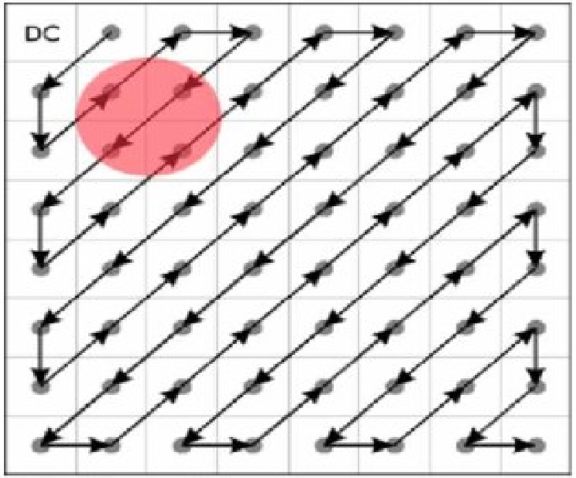
$C(m), C(n) = 1/\sqrt{2}$ **pro** $m,n = 0$

jinak

$C(m), C(n) = 1$

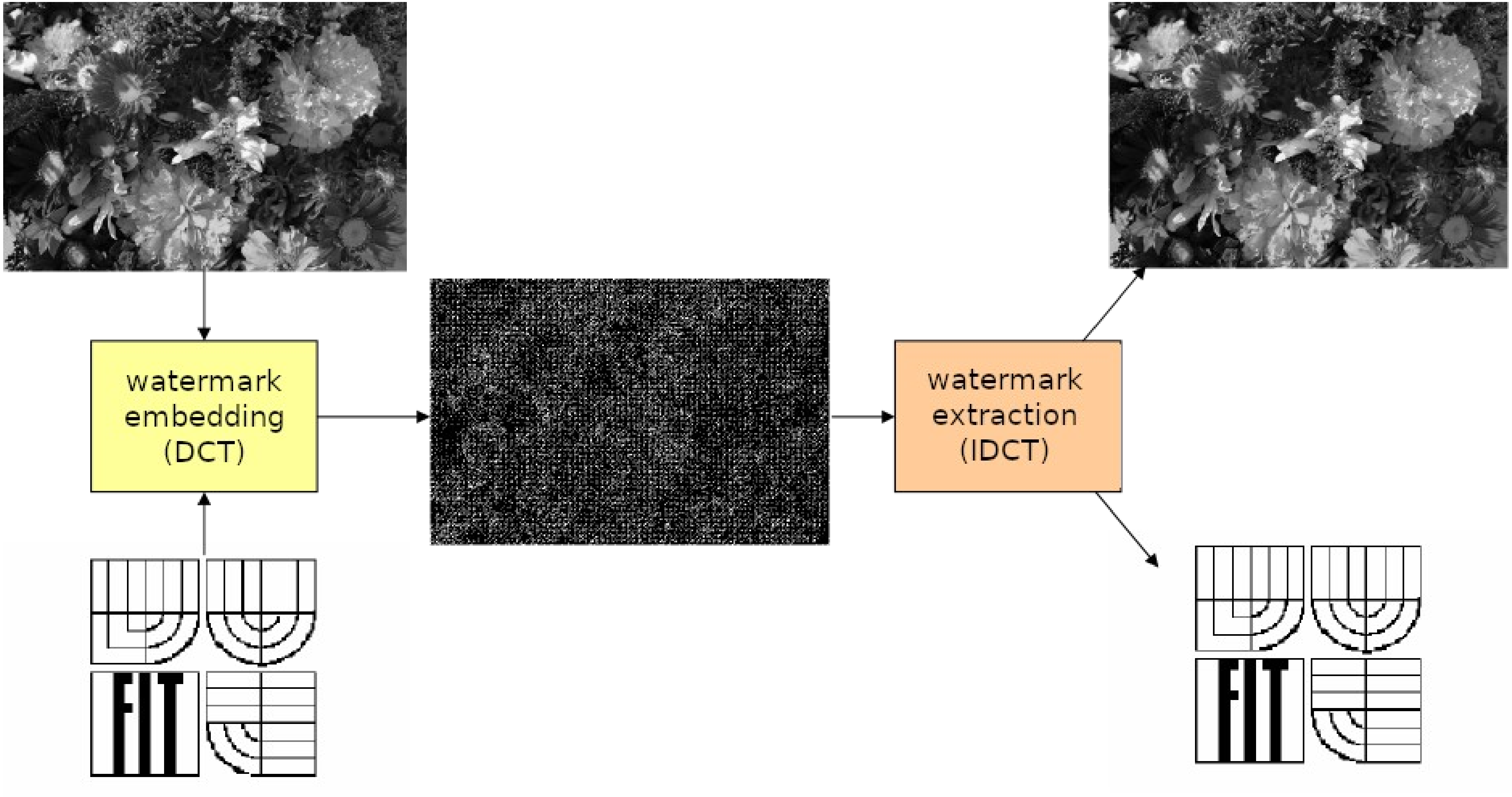
Embedding of watermark using DCT coefficients





- Changing the lower frequency coefficients is more robust, but it affects more image quality.
- DC coefficient is unchanged, the change would be more visible, the watermark is resistant to change of brightness.
- Several coefficients are selected. Each bit of a watermark is encoded into single coefficient.
- The watermark can be repeated in a block. It is more resistant to image cropping.

Embedding of watermark using DCT coefficients

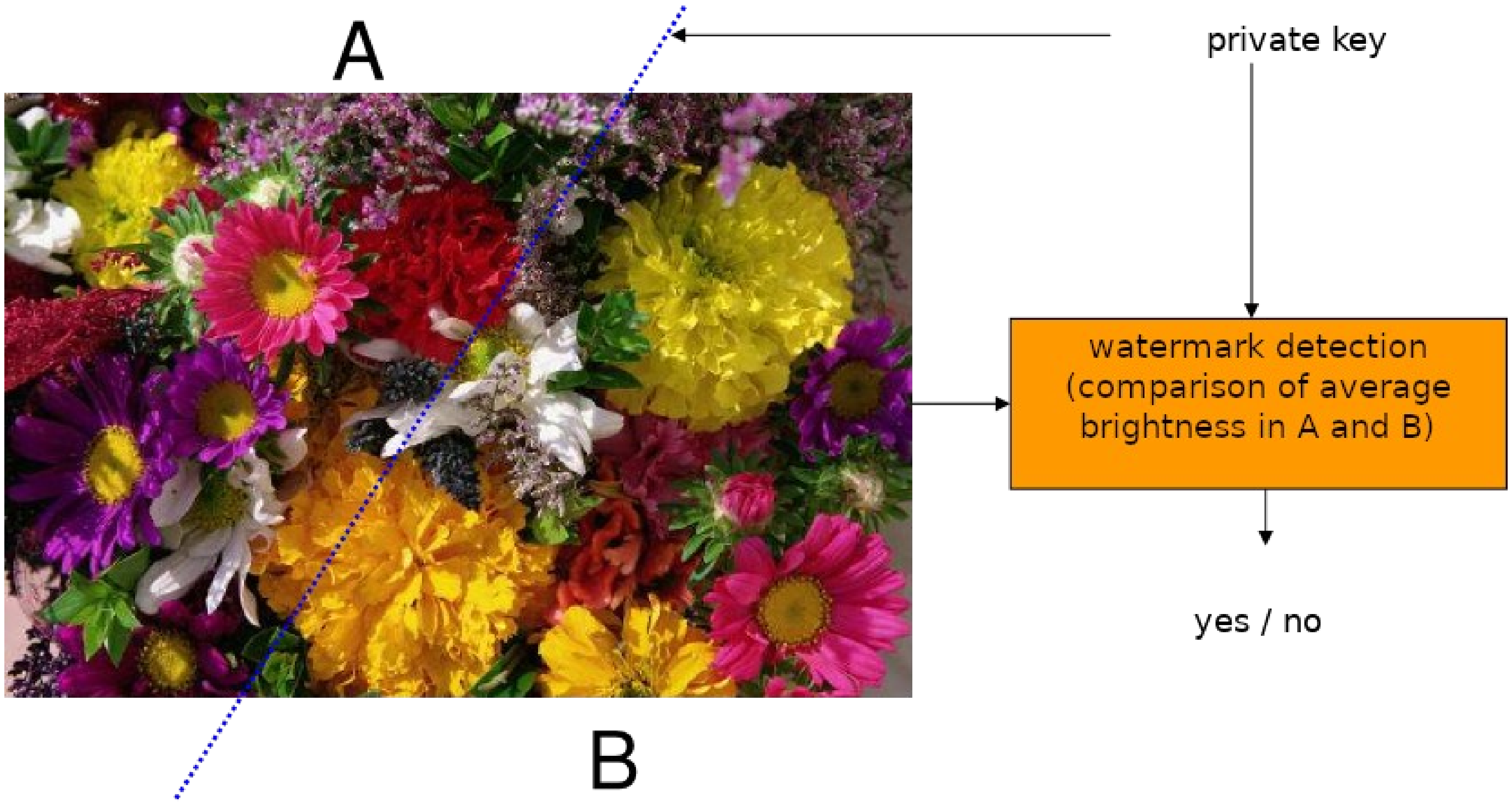


Disjoint sets

- How to embed watermark
  - Split image into two approximately equal large sets of pixels A, B.
  - Selected pixels are private key.
  - The pixel intensity in set A is increased by k.
  - The pixel intensity in set A is decreased by k.
  - Change must be rather small to be invisible in output image.

- Detection of watermark
  - Split image according to private key.
  - Compute average intensity in both sets.
  - If the watermark is included, the intensity difference is approximately 2k. Otherwise, almost 0.

Disjoint sets

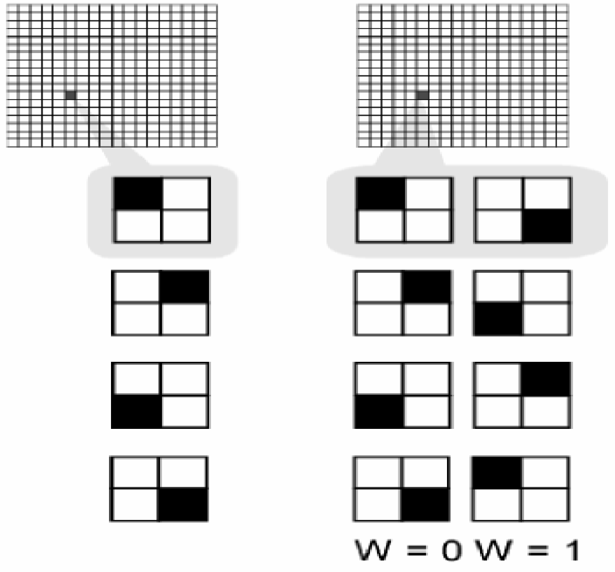


Use of MD5

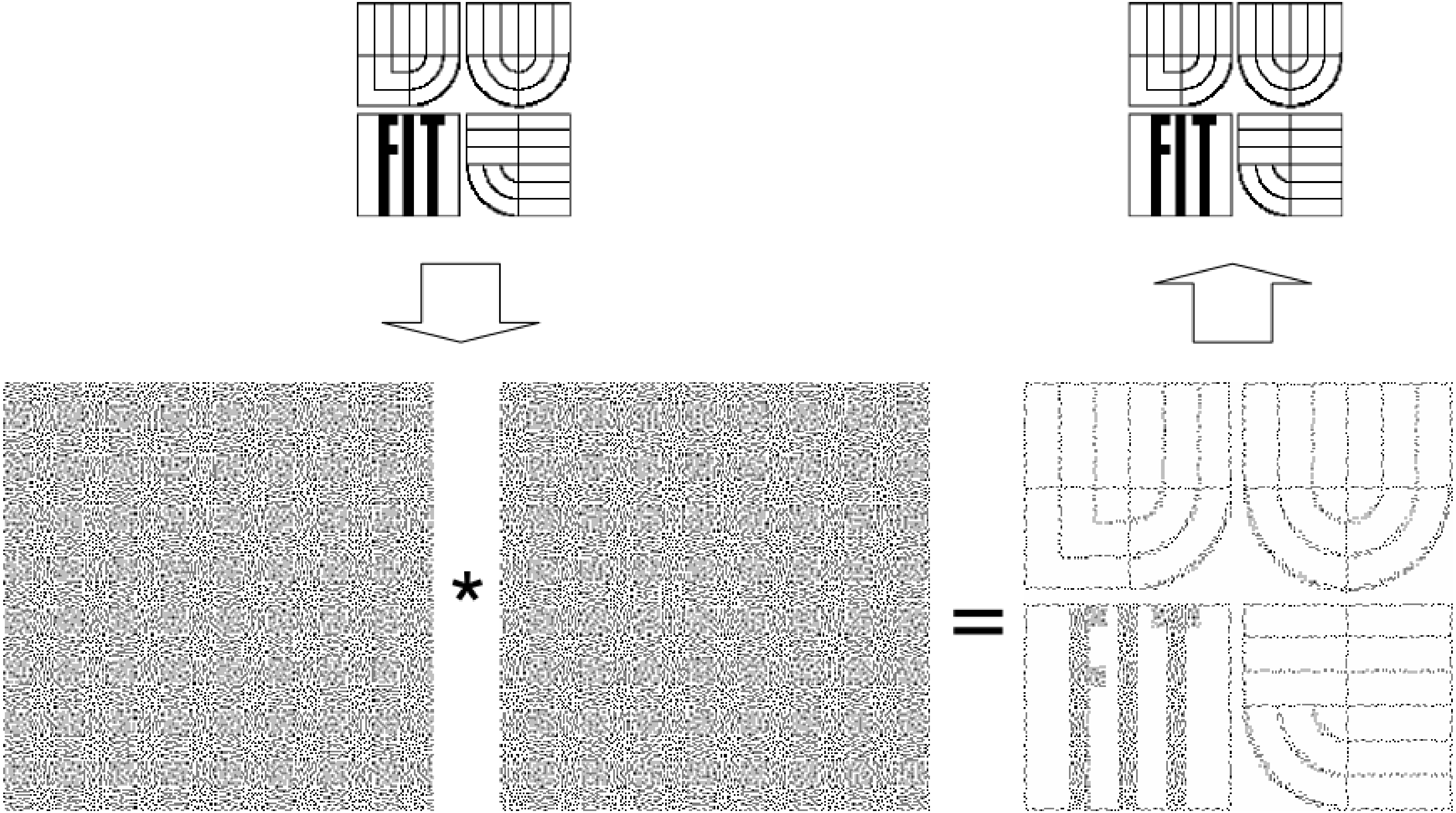
- Embeding of watermark
  - Image is diveded into block of 8x8 pixels.
  - Least significant bits are removed.
  - The rest of the bits are concatenated with a parameter dependent on image size and private key.
  - The MD5 checksum of it is combined with the binary image of the watermark.
  - Result is inserted into image as least significant bits.
- It allows pixel-wise detection of modifications.
- Watermark could be used to prove authenticity.

Visual crypthography

- Watermarking of half-toned images.
- The message is stored in consecutive binary images.
- When viewing a single image, the watermark is not visible.
- The watermark is visible after applying the AND logical operation to the single pixel pixels.
- Watermark encoding
  - Source pixels 4x larger.
  - The left image is randomly encoded by one of the four op
  - Right image encoded by quadruple based on source pixe
- Use for steganography.
- U.S. Patent No. 5,488,664.



Visual crypthography



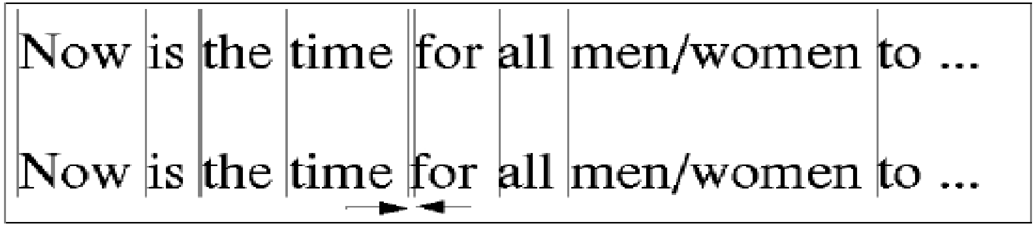
Dear George,
Greetings to all at Oxford. Many thanks for your
letter and for the summer examination package.
All entry forms and fees forms should be ready
for final dispatch to the syndicate by Friday
20th or at the latest I am told by the 21st.
Admin has improved here though there is room
for improvement still; just give us all two or three
more years and we will really show you! Please
don't let these wretched 16+ proposals destroy
your basic O and A pattern. Certainly this
sort of change, if implemented immediately,
would bring chaos.

Sincerely yours,

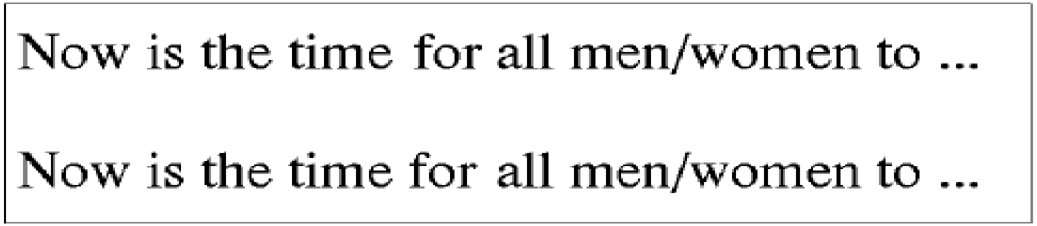
Watermarks in text

- Parameters of text rate are changed
- For example:
  - Line could be moved up or down
  - Spaces between words could have different width
  - Encoding into font. Extending width of horizontal line in “t” character.

a)

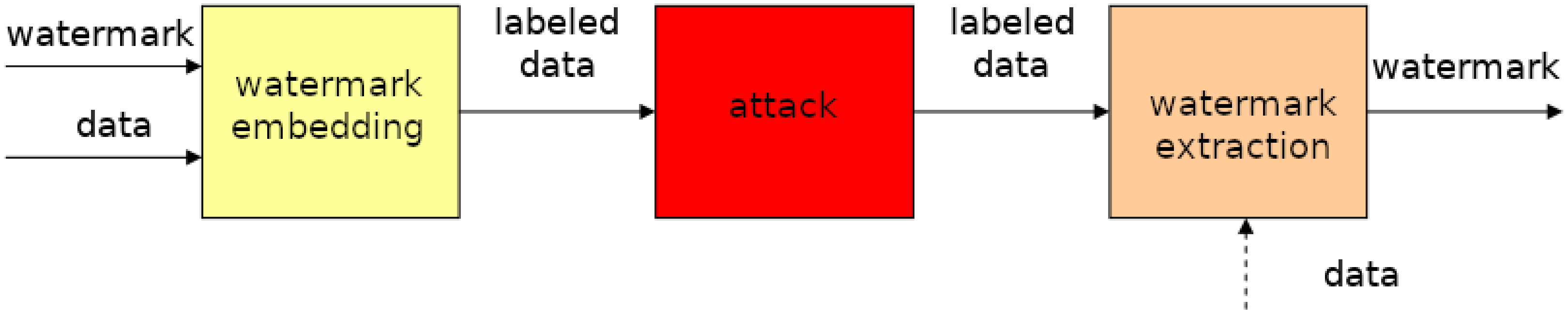


b)



Attacks

- Any action that accidentally or deliberately damages encoded information.
- Reduction of the effectiveness or complete removal of the watermark while maintaining the utility value of the work.
- Attacks as a communication channel.
- Robustness - the embedded information can not be corrupted or removed without the data being rendered rendered unusable. Difficult to evaluate.



Attacks

- Methods based on knowledge of watermark embedding algorithm
  - Inserting additional noise into each pixel.
  - Repeated application of common operations such as compression, filtering, scaling, rotation, printing, and scanning.
- Splitting the image into blocks that are assembled until the presentation - the blocks are so small that they can not find a complete watermark.
- The "brute force" method - usable if the secret key is too short - can be used to test all options.
- Insert another watermark
  - The attacker adds his watermark of the same weight to the work and declares the work as his own.
  - Both watermarks can be detected and it is not possible to identify the original owner.
  - Therefore, the original watermark need not be modified or removed.

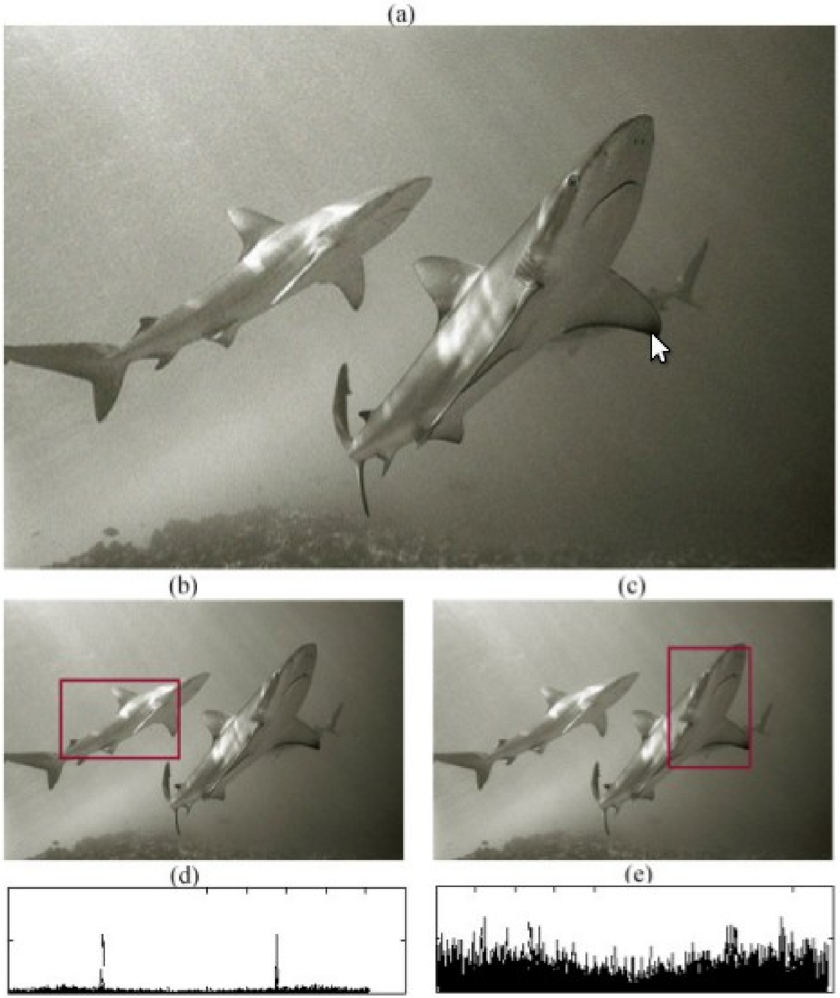
Attacks

- Conspiracy of "users"
  - Several customers buy the same work eg a picture with a unique watermark.
  - By averaging individual copies, the watermark can be weakened or removed altogether.

Detection of hidden watermark

- interpolation and resampling,
- inconsistencies in chromatic aberration,
- noise inconsistencies,
- double JPEG compression,
- inconsistencies in color filter array (CFA) interpolated images,
- inconsistencies in lighting.

Forensic analysis of image



Summary

- Encryption is suitable for data transfer, but it does not provide for the subsequent protection of the work.
- Watermarks are a mechanism to protect data even after decryption.
- Cryptographic algorithms can usually be mathematically proven. It is dificult to measure robustness and invisibility in watermarks.
- The attack on cryptography means finding secrets; attacks on watermarks can be just modification or removal of information.
- Watermarks could be used for copyright protection, copy protection, integrity of work detection, steganography, ...
- Watermark types - visible X invisible (fragile, robust).

- https://github.com/fgrimme/Matroschka
- http://enigmatalk.com/ https://www.youtube.com/watch?v=ikA4Sqq7BZQ
- http://incoherency.co.uk/image-steganography/#
- https://sourceforge.net/projects/image-steg/
- http://download.cnet.com/Xiao-Steganography/3000-2092_4-10541494.html
- http://imagesteganography.codeplex.com/
- http://manytools.org/hacker-tools/steganography-encode-text-into-image/
- https://github.com/dipanjanS/Digital-image-steganography
- https://github.com/JarrodCTaylor/steganopy
- https://github.com/ragibson/Steganography
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